

High Level Architecture - Ignitho AI Suite ENTERPRISE SECURITY & COMPLIANCE PERIMETER (SOC2 Type II | ISO 27001 | HIPAA Compliant | Zero-Trust Mesh | mTLS 1.3)

1. STAKEHOLDERS & INGRESS TIER



Business Users & Engineers

- Executive Dashboards
- Pipeline Studio Operators



Enterprise Ingress & WAF Gateway

- OAuth2 / SSO Authentication Provider
- Rate Limiting & DDoS Shield
- TLS 1.3 Ingress Termination

2. KUBERNETES ORCHESTRATION PLATFORM (EKS / Helm / Terraform Managed)

A. Frontend Experience Microservices



Next.js App Router Server



React Flow Studio & Tailwind CSS
ShadCN Visual Canvas Engine



Client PII Anonymizer Node
Presidio & Regex Sanitizer

B. Middleware & Orchestration Engine



Node.js API Gateway & Controller
JWT Validation & Rate Limits



Stateful DAG Workflow Executor
Parallel Pipeline Processing



RabbitMQ Event Bus Cluster
Async Queues & Backpressure Control

C. Python AI Agent & RAG Intelligence Suite



FastAPI Microservices Engine (Python 3.11)
High-Performance Async API Endpoints



Multi-Agent Execution Framework
• Data Ingestion & ETL Agents • Audit & Quality Agents
• Clinical Trial Matchers • Financial Invoice Processors



Enterprise RAG & Knowledge Synthesizer
LangChain, Vector Search & Context Assembly



AI Fortress Firewall & Token Cost Router
Blocks Injection Attacks, Hallucinations & Controls Cost



Secrets & Key Management
(HashiCorp Vault / AWS Secrets Manager)

- Encrypted ENV at Rest
- Private API Credentials
- Master KMS Encryption Keys



Air-Gapped Local Model Host (Ollama Container) - Used for Offline Confidential In-House Testing

3. ENTERPRISE DATASETS, STORAGE & FOUNDATION MODEL TIER (Encrypted at Rest - AES 256)



MongoDB Cluster

- User Profiles & Audit Logs
- WiredTiger AES-256 Storage



Upstash Redis Cache

- Distributed State & Rate Limits
- Low-Latency Session Cache



Qdrant Vector Cluster

- Enterprise Knowledge Base
- High-Dimensional Embeddings



Amazon S3 Bucket

- Unstructured Documents
- Enforced SSE-KMS Encryption



AWS Bedrock Foundation Models

- Claude, Llama & Titan Models
- PrivateLink Network Isolation



Enterprise DevOps & IaC Lifecycle: Terraform Infrastructure Provisioning | Helm Chart Deployments | Docker Containerization | GitOps Versioning

